

Question ID: fe287f7e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

To repair a refrigerator, a technician charges **\$60** per hour for labor plus **\$120** for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

Answer

- A. $f(x) = x + 120$
- B. $f(x) = 60x$
- C. $f(x) = 60x + 120$
- D. $f(x) = 60x - 120$

Correct Answer: C

Rationale

Choice C is correct. It's given that the technician charges **\$60** per hour for labor. Therefore, if the job takes x hours, the technician will charge $\left(\frac{\$60}{1 \text{ hour}}\right)(x \text{ hours})$, or **\$60x**, for labor. It's also given that the technician charges **\$120** for parts. Therefore, $f(x) = 60x + 120$ represents the total amount, in dollars, the technician will charge for this job if it takes x hours.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This function represents the total amount, in dollars, the technician charges for labor only, not the total amount charged for labor and parts.

Choice D is incorrect. This function represents the total amount, in dollars, the technician would charge if the charge for parts were subtracted from, rather than added to, the charge for labor.